

Notes Power Model D-MIMO

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I. CENTRAL MIMO POWER MODEL

We adopt the parametric power model for upper mid-band (FR3) base stations of [1]. The base station (BS) is equipped with M_{ant} antennas, an equal number of power amplifiers (PAs, used in the downlink) and low-noise amplifiers (LNAs, used in the uplink), and M_{RF} radio-frequency (RF) chains. It serves K single-antenna users over a bandwidth B at carrier frequency f_c using OFDM and space-division multiple access. Setting $M_{\text{RF}} = M_{\text{ant}}$ selects fully-digital beamforming; $M_{\text{RF}} < M_{\text{ant}}$ selects hybrid partially-connected beamforming with $M_{\text{ant}}/M_{\text{RF}}$ phase shifters per RF chain.

A. Model structure

The total consumed power is split into three hardware blocks; digital signal processing, analog signal processing, and the PAs, each divided by its supply-and-cooling efficiency $\eta_{c,\text{sc}} \in (0, 1]$:

$$P_{\text{cons}} = \frac{\overline{P_{\text{dig}}}}{\eta_{\text{dig,sc}}} + \frac{\overline{P_{\text{ana}}}}{\eta_{\text{ana,sc}}} + \frac{\overline{P_{\text{PA}}}}{\eta_{\text{PA,sc}}}. \quad (1)$$

Each block is built from the working-mode (active) consumption of its subcomponents, averaged over the operating modes within a frame. We first define this frame averaging (Section I-B) and then reuse it as a function for every component (Sections I-C to I-E).

B. Frame structure and operating-mode averaging

A frame of duration T_f contains a downlink (DL) and an uplink (UL) phase under time division duplexing (TDD), with duration ratios $\tau_{\text{DL}}, \tau_{\text{UL}} \in [0, 1]$ and $\tau_{\text{UL}} = 1 - \tau_{\text{DL}}$. Within each direction $i \in \{\text{DL}, \text{UL}\}$ the time is further subdivided into three operating modes, plus an idle mode when the direction is inactive:

- *reference signalling*, for a time fraction $\tau_i \tau_{i,\text{sig}}$;
- *data transmission/reception*, for a fraction $\tau_i (1 - \tau_{i,\text{sig}}) \frac{N_{a,i}}{N_i}$;
- *micro-sleep* (DTX), for a fraction $\tau_i (1 - \tau_{i,\text{sig}}) \left(1 - \frac{N_{a,i}}{N_i}\right)$;
- *idle*, for a fraction $(1 - \tau_i)$,

where N_i is the number of OFDM symbols (time slots) of direction i and $N_{a,i} \leq N_i$ the subset that are *active* (carrying data). The *active-resource time ratio* $N_{a,i}/N_i$ is what sets the data-versus-micro-sleep split: a slot is either active or not, regardless of how many users share it. with $\tau_{i,\text{sig}} \in [0, 1]$ the signalling fraction of direction i .

thanks

a) *Average physical resource load*: The data-mode weight is more conveniently expressed through the *average physical resource load* $\bar{x}_i$, which we now relate to the time ratio $N_{a,i}/N_i$. Let K_{max} be the maximum number of users that can be served (the number of spatial streams) and Q_{max} the number of available data subcarriers per active OFDM symbol. The instantaneous load of user k in time slot n is the ratio of utilised to available data subcarriers,

$$\ell_i[k, n] = \frac{Q_i[k, n]}{Q_{\text{max}}}, \quad i \in \{\text{DL}, \text{UL}\}, \quad (2)$$

for $k = 1, \dots, K_{\text{max}}$ and $n = 1, \dots, N_i$. Averaging over all K_{max} potential users and N_i slots gives the average physical resource load

$$\bar{x}_i \triangleq \bar{\ell}_i = \frac{1}{N_i} \sum_{n=1}^{N_i} \left(\frac{1}{K_{\text{max}}} \sum_{k=1}^{K_{\text{max}}} \frac{Q_i[k, n]}{Q_{\text{max}}} \right). \quad (3)$$

Under space-division multiple access (no OFDMA), an active user in an active slot is allocated *all* data subcarriers, $Q_i[k, n] = Q_{\text{max}}$. Indexing the K_i active users as $\{1, \dots, K_i\}$ and the $N_{a,i}$ active slots as $\{1, \dots, N_{a,i}\}$, each active-user/active-slot pair contributes 1 to (3) and every other pair 0; the double sum equals $K_i N_{a,i}$ and

$$\bar{x}_i = \frac{N_{a,i}}{N_i} \frac{K_i}{K_{\text{max}}}. \quad (4)$$

b) *Equivalence and modelling assumption*: The duty-cycle split in the frame structure is governed by the active-resource time ratio $N_{a,i}/N_i$, whereas the power model uses the average load $\bar{x}_i$ as the data-mode weight. Comparing (4) with the data fraction $\tau_i (1 - \tau_{i,\text{sig}}) N_{a,i}/N_i$, the two agree,

$$\bar{x}_i = \frac{N_{a,i}}{N_i} \iff \frac{K_i}{K_{\text{max}}} = 1 \iff K_i = K_{\text{max}}, \quad (5)$$

i.e. only when every spatial stream is scheduled. When the maximum number of users is active the average load is simply the ratio of active to total slots; for $K_i < K_{\text{max}}$ the average load (4) is the true active fraction scaled by $K_i/K_{\text{max}} < 1$. The number of active users K_i separately sets the transmit power and the achievable rate, but not whether a slot is active. We use $\bar{x}_i$ as the data-mode weight throughout (and $1 - \bar{x}_i$ as the micro-sleep/DTX fraction); this is therefore *exact* in the full-spatial-load regime $K_i = K_{\text{max}}$ evaluated here and an approximation otherwise.

In micro-sleep and idle a component drops to a fraction of its working consumption, set by the reduction factors $\delta_c^\mu, \delta_c^0 \in [0, 1]$. For a component c with working power P^a during data, P^s during signalling, and base power P^0 scaled in the reduced

modes, the frame-averaged consumption of direction i is the time-weighted sum

$$\begin{aligned} \mathcal{A}_i^c(P^a, P^s, P^0) &\triangleq \bar{x}_i \tau_i (1 - \tau_{i,\text{sig}}) P^a + \tau_i \tau_{i,\text{sig}} P^s \\ &\quad + (1 - \bar{x}_i) \tau_i (1 - \tau_{i,\text{sig}}) \delta_c^\mu P^0 \\ &\quad + (1 - \tau_i) \delta_c^0 P^0. \end{aligned} \quad (6)$$

For the digital and analog blocks the consumption is identical in all working modes, $P^a = P^s = P^0 = P$, and we use the shorthand $\mathcal{A}_i^c(P) \triangleq \mathcal{A}_i^c(P, P, P)$. Only the PA distinguishes the three levels (Section I-C).

Each direction's hardware is averaged over the *whole* frame, and the two directions are accounted for separately, e.g. for the digital block $\overline{P_{\text{dig}}} = \mathcal{A}_{\text{DL}}^{\text{dig}}(P_{\text{dig},\text{DL}}) + \mathcal{A}_{\text{UL}}^{\text{dig}}(P_{\text{dig},\text{UL}})$. The idle term $(1 - \tau_i) \delta_c^0 P^0$ is therefore the reduced consumption of direction i 's chain while that direction is *not* active, i.e. while the opposite direction occupies the frame. For $i = \text{DL}$ this fraction equals τ_{UL} only because the gap-free TDD frame satisfies $\tau_{\text{UL}} = 1 - \tau_{\text{DL}}$; it represents the idle DL chain, not the active UL processing, which is captured by the separate $\mathcal{A}_{\text{UL}}$ term.

Remark 1 (Load-dependent split). *Only the data term in (6) scales with the load $\bar{x}_i$, while the micro-sleep term scales with $(1 - \bar{x}_i)$. Collecting the terms proportional to $\bar{x}_i$ separates each averaged consumption into a load-dependent part and a load-independent part, which is used in the analysis to attribute consumption to the delivered traffic.*

C. Power amplifiers

The PAs transmit only in the DL. With a per-PA output power $P_a = P_{T,\text{DL}}/M_{\text{ant}}$, where $P_{T,\text{DL}}$ is the total DL transmit power, the instantaneous consumption of one PA is modelled as

$$P_{\text{PA}}(p) = \frac{1}{\eta_{\text{PA},\text{max}}} [\xi P_{\text{max}} + (1 - \xi) P_{\text{max}}^{1-\alpha} p^\alpha], \quad (7)$$

where p is the output power, P_{max} the maximum output power (at a fixed back-off from saturation), $\eta_{\text{PA},\text{max}}$ the maximum PA efficiency, $\xi \in [0, 1]$ the static-to-dynamic consumption ratio, and $\alpha \in [0, 1]$ the efficiency exponent. Averaging over the DL frame and summing over the M_{ant} PAs gives

$$\overline{P_{\text{PA}}} = M_{\text{ant}} \mathcal{A}_{\text{DL}}^{\text{PA}}(P_{\text{PA}}(P_a), P_{\text{PA}}(\zeta_{\text{sig}} P_{\text{max}}), P_{\text{PA}}(0)), \quad (8)$$

where $\zeta_{\text{sig}} \in (0, 1]$ is the fraction of P_{max} used during signalling, and the zero-output consumption $P_{\text{PA}}(0)$ is the base level scaled in micro-sleep and idle.

D. Analog signal processing

The analog block comprises, per RF chain, the DACs (DL) or ADCs (UL), RF filters, mixers, M_{PS} phase shifters, and the LNAs (UL); the local oscillator (LO) is shared across chains. With $M_{\text{PS}} = M_{\text{ant}}/M_{\text{RF}}$ for hybrid beamforming ($M_{\text{PS}} = 0$

for fully-digital), the working-mode consumption per direction is

$$P_{\text{ana},\text{DL}} = P_{\text{LO}} + M_{\text{RF}}(2P_{\text{DAC}} + 2P_{\text{filt},\text{RF}} + 2P_{\text{mix}} + M_{\text{PS}}P_{\text{PS}}), \quad (9)$$

$$P_{\text{ana},\text{UL}} = P_{\text{LO}} + M_{\text{RF}}(2P_{\text{ADC}} + 2P_{\text{filt},\text{RF}} + 2P_{\text{mix}} + M_{\text{PS}}P_{\text{PS}} + \frac{M_{\text{ant}}}{M_{\text{RF}}}P_{\text{LNA}}), \quad (10)$$

with the factor 2 accounting for the in-phase and quadrature branches. The subcomponents are modelled as

$$P_{\text{DAC}} = \Xi_{\text{DAC},1} 2^{b_{\text{DAC}}} + \Xi_{\text{DAC},2} b_{\text{DAC}} f_{\text{DAC}}, \quad (11)$$

$$P_{\text{ADC}} = W_{\text{ADC}} f_{\text{ADC}} 2^{b_{\text{ADC}}}, \quad (12)$$

$$P_{\text{mix}} = \Xi_{\text{mix}} f_c, \quad (13)$$

$$P_{\text{PS}} = \Xi_{\text{PS}} B, \quad (14)$$

$$P_{\text{LNA}} = \Xi_{\text{LNA}} B, \quad (15)$$

where $b_{\text{DAC}}, b_{\text{ADC}}$ are the effective resolutions, $f_{\text{DAC}}, f_{\text{ADC}}$ the converter sampling rates, W_{ADC} the ADC Walden figure-of-merit, and $\Xi_\bullet$ the respective figures-of-merit. The frame-averaged analog consumption is

$$\overline{P_{\text{ana}}} = \mathcal{A}_{\text{DL}}^{\text{ana}}(P_{\text{ana},\text{DL}}) + \mathcal{A}_{\text{UL}}^{\text{ana}}(P_{\text{ana},\text{UL}}) + P_{\text{ana},\text{misc}}. \quad (16)$$

E. Digital signal processing

In the DL the digital block runs the channel encoder, MIMO precoder, IFFT, DPD, and baseband (BB) filter; in the UL the channel decoder, MIMO combiner, FFT, and BB filter. Each subcomponent consumption is a static (signal-independent) term plus a dynamic term equal to its complex giga-operations-per-second (GOPS) Ξ divided by the computational efficiency η (in GOPS/W) of the chosen FPGA/ASIC implementation: **TODO red channel estimation cost?**

$$P_{\text{enc}} = P_{\text{enc},s} + \frac{f_{s,\text{I}}}{\eta_{\text{enc}}} \frac{14}{24} \frac{R_{\text{DL}}}{\tilde{B}}, \quad (17)$$

$$P_{\text{dec}} = P_{\text{dec},s} + \frac{f_{s,\text{I}}}{\eta_{\text{dec}}} \frac{175}{6} \frac{R_{\text{UL}}}{\tilde{B}}, \quad (18)$$

$$P_{\text{pre}} = \bar{M}_{\text{RF}} P_s + \frac{f_{s,\text{I}}}{\eta_{\text{pre}}} M_{\text{RF}} \Xi_{\text{pre}}, \quad (19)$$

$$P_{\text{comb}} = \bar{M}_{\text{RF}} P_s + \frac{f_{s,\text{I}}}{\eta_{\text{comb}}} M_{\text{RF}} 2K, \quad (20)$$

$$P_{\text{IFFT}} = P_{\text{FFT}} = P_{\text{IFFT},s} + \frac{f_{s,\text{II}}}{\eta_{\text{IFFT}}} M_{\text{RF}} \frac{3}{2} \log_2 Q_{\text{IFFT}}, \quad (21)$$

$$P_{\text{DPD}} = P_{\text{DPD},s} + \frac{f_{s,\text{II}}}{\eta_{\text{DPD}}} M_{\text{RF}} \Xi_{\text{DPD}}, \quad (22)$$

$$P_{\text{filt},\text{BB}} = \bar{M}_{\text{RF}} P_s + \frac{f_{s,\text{II}}}{\eta_{\text{BB}}} M_{\text{RF}} \Xi_{\text{BB}}, \quad (23)$$

where $R_{\text{DL}}, R_{\text{UL}}$ are the delivered DL/UL sum rates, $\tilde{B}$ the effective bandwidth, $M_{\text{RF}} = \lceil M_{\text{RF}}/32 \rceil$ the number of FPGAs (one per 32 chains, sharing the precoder/combiner/BB-filter static power P_s), and Q_{IFFT} the IFFT/FFT size. The two sampling rates are the constellation rate $f_{s,\text{I}} = \mu B$ (for encoding/decoding and MIMO processing) and the oversampled rate

$f_{s,II} = Q_{\text{IFFT}} \Delta f$ (for the OFDM, DPD, and BB operations), with $\mu \in [0, 1]$ and subcarrier spacing Δf . For the zero-forcing MIMO processor exploiting channel reciprocity, the precoder GOPS is **TODO this changes for cell free depending on the precoder used**

$$\Xi_{\text{pre}} = 2K + \frac{1}{v_{\text{coh}}} \left(\frac{K^3}{3M_{\text{RF}}} + 3K^2 + K \right), \quad (24)$$

with v_{coh} the number of samples per channel coherence block over which the precoder is reused; the combiner only applies the precoding matrix, giving $\Xi_{\text{comb}} = 2K$. The active digital consumption per direction is the sum of its subcomponents,

$$P_{\text{dig,DL}} = P_{\text{enc}} + P_{\text{pre}} + P_{\text{IFFT}} + P_{\text{DPD}} + P_{\text{filt,BB}}, \quad (25)$$

$$P_{\text{dig,UL}} = P_{\text{dec}} + P_{\text{comb}} + P_{\text{IFFT}} + P_{\text{filt,BB}}, \quad (26)$$

and the frame-averaged digital consumption is

$$\overline{P_{\text{dig}}} = \mathcal{A}_{\text{DL}}^{\text{dig}}(P_{\text{dig,DL}}) + \mathcal{A}_{\text{UL}}^{\text{dig}}(P_{\text{dig,UL}}). \quad (27)$$

F. Delivered rate and energy efficiency

The achievable ergodic sum rate of direction i , which drives the encoder/decoder consumption in (17), follows from the per-user SINRs under zero-forcing beamforming as

$$R_i = \bar{x}_i \tau_i (1 - \tau_{i,\text{sig}}) \frac{\tilde{B}}{q_i} \mathbb{E} \left\{ \sum_{\nu=1}^{q_i} \sum_{k=1}^K \log_2(1 + \text{SINR}_{i,k}[\nu]) \right\}, \quad (28)$$

where the pre-log collects the load, duration, signalling-overhead, and effective-bandwidth-per-subcarrier factors. The base-station energy efficiency then follows as the delivered rate per consumed watt,

$$\text{EE} = \frac{R_{\text{DL}} + R_{\text{UL}}}{P_{\text{cons}}} \quad [\text{bit/J}]. \quad (29)$$

II. DISTRIBUTED MASSIVE MIMO DOWNLINK SYSTEM MODEL

We consider the downlink of a distributed (cell-free) massive multiple input multiple output (MIMO) network in which L access points (APs), each equipped with M antennas, jointly serve K single-antenna users over the same time-frequency resources. The APs are connected through fronthaul links to a central processing unit (CPU) that encodes the data and computes the precoders, so that the LM antennas act as a single distributed array. Transmission uses OFDM with Q data subcarriers, indexed by $q = 1, \dots, Q$; the model below is stated per subcarrier and the same processing is applied on each of them.

Let $\mathbf{h}_{kl}[q] \in \mathbb{C}^M$ denote the channel from AP l to user k on subcarrier q . Stacking the contributions of all APs gives the collective channel

$$\mathbf{h}_k[q] = \begin{bmatrix} \mathbf{h}_{k1}[q] \\ \vdots \\ \mathbf{h}_{kL}[q] \end{bmatrix} \in \mathbb{C}^{LM}. \quad (30)$$

The channel realizations are drawn from the 3GPP TR 38.901 urban-microcell (UMi) geometry-based stochastic channel

model [2], using the implementation provided by the Sionna link-level simulator [3]. Each AP is modelled as a 38.901 base station carrying an M -element half-wavelength uniform linear array (ULA) of omnidirectional elements and each user as a single-antenna terminal. Every AP-user link is assigned a line-of-sight or non-line-of-sight condition according to the distance-dependent LOS probability of the UMi scenario, and its distance-dependent path loss, log-normal shadow fading, and small-scale multipath then determine $\mathbf{h}_{kl}[q]$.

Because the model is specified directly in terms of channel realizations rather than through a closed-form spatial correlation matrix, the large-scale fading coefficient β_{kl} between AP l and user k , which drives the downlink power control of Section II-A, is obtained from the realizations as the average per-antenna channel gain,

$$\beta_{kl} = \frac{1}{M} \mathbb{E} \left(\|\mathbf{h}_{kl}[q]\|^2 \right), \quad (31)$$

which is estimated by averaging $|\mathbf{h}_{kl}[q]_m|^2$ over the M antennas of AP l and the Q subcarriers. It captures the combined effect of path loss and shadow fading (and, on a LOS link, of the specular component) and so summarizes the slow, position-dependent part of the channel that the power allocation acts on.

The CPU maps the data symbol $s_k[q] \in \mathbb{C}$ of user k , with unit power $\mathbb{E}(|s_k[q]|^2) = 1$ and mutually independent across users, onto the M antennas of each AP. AP l transmits

$$\mathbf{x}_l[q] = \sum_{k=1}^K \mathbf{w}_{kl}[q] s_k[q] = \mathbf{W}_l[q] \mathbf{s}[q], \quad (32)$$

where $\mathbf{w}_{kl}[q] \in \mathbb{C}^M$ is the precoding vector that AP l applies to user k , $\mathbf{s}[q] = [s_1[q] \cdots s_K[q]]^\top \in \mathbb{C}^K$ collects the data symbols, and $\mathbf{W}_l[q] = [\mathbf{w}_{l1}[q] \cdots \mathbf{w}_{lK}[q]] \in \mathbb{C}^{M \times K}$ is the precoding matrix of AP l . Collecting the per-AP precoders of a given user into $\mathbf{w}_k[q] = [\mathbf{w}_{k1}[q]^\top \cdots \mathbf{w}_{kL}[q]^\top]^\top \in \mathbb{C}^{LM}$ yields the network-wide precoding matrix

$$\mathbf{W}[q] = [\mathbf{w}_1[q] \cdots \mathbf{w}_K[q]] \in \mathbb{C}^{LM \times K}, \quad (33)$$

whose l -th block of M rows equals $\mathbf{W}_l[q]$, so that the stacked transmit vector is $\mathbf{x}[q] = [\mathbf{x}_1[q]^\top \cdots \mathbf{x}_L[q]^\top]^\top = \mathbf{W}[q] \mathbf{s}[q] \in \mathbb{C}^{LM}$.

The signal received by user k on subcarrier q is $y_k[q] = \mathbf{h}_k[q]^H \mathbf{x}[q] + n_k[q]$, which, using (33), separates into the desired term, the multi-user interference, and the noise,

$$y_k[q] = \underbrace{\mathbf{h}_k[q]^H \mathbf{w}_k[q] s_k[q]}_{\text{desired signal}} + \underbrace{\sum_{\substack{i=1 \\ i \neq k}}^K \mathbf{h}_k[q]^H \mathbf{w}_i[q] s_i[q]}_{\text{multi-user interference}} + \underbrace{n_k[q]}_{\text{noise}}, \quad (34)$$

where $n_k[q] \sim \mathcal{CN}(0, \sigma^2)$ is the receiver noise with power σ^2 , independent across users and subcarriers and independent of the channels and data. The scalar $\mathbf{h}_k[q]^H \mathbf{w}_k[q]$ is the effective channel seen by user k : the distributed precoder turns the LM -antenna channel into an equivalent single-antenna link.

Treating the multi-user interference in (34) as noise, the effective downlink signal-to-interference-plus-noise ratio (SINR) of user k on subcarrier q is

$$\text{SINR}_k^{\text{dl}}[q] = \frac{|\mathbf{h}_k[q]^H \mathbf{w}_k[q]|^2}{\sum_{\substack{i=1 \\ i \neq k}}^K |\mathbf{h}_i[q]^H \mathbf{w}_i[q]|^2 + \sigma^2}. \quad (35)$$

With the effective channel known at the receiver, an achievable rate for user k on subcarrier q is $\log_2(1 + \text{SINR}_k^{\text{dl}}[q])$, and the delivered downlink sum rate R_{DL} follows by inserting these per-user, per-subcarrier SINRs into the ergodic rate expression of Section I-F ((28)), which in turn drives the encoder and MIMO processing consumption of the power model.

A. Precoder and Power Control Designs

Unlike co-located massive MIMO, where a single sum-power budget applies to the whole array, each AP is a separate transmitter and is subject to its own power constraint. Since the data symbols are independent and unit-power, the average power radiated by AP l across the OFDM block is

$$\sum_{q=1}^Q \mathbb{E}(\|\mathbf{x}_l[q]\|^2) = \sum_{q=1}^Q \sum_{k=1}^K \mathbb{E}(\|\mathbf{w}_{kl}[q]\|^2) \leq \rho_{\max}, \quad (36)$$

$$l = 1, \dots, L, \quad (37)$$

where $\rho_{\max}$ is the maximum downlink transmit power per AP, assumed equal for all APs, and the expectation is taken over the data and channel realizations.

It is convenient to split each precoder into a transmit direction and a power control coefficient,

$$\mathbf{w}_k[q] = \sqrt{\rho_k} \frac{\bar{\mathbf{w}}_k[q]}{\sqrt{\sum_{q'=1}^Q \mathbb{E}(\|\bar{\mathbf{w}}_k[q']\|^2)}}, \quad (38)$$

where $\bar{\mathbf{w}}_k[q] \in \mathbb{C}^{LM}$ is an arbitrarily scaled vector that fixes the spatial direction of the transmission to user k and $\rho_k \geq 0$ is the downlink power allocated to that user across the OFDM block, so that $\sum_{q=1}^Q \mathbb{E}(\|\mathbf{w}_k[q]\|^2) = \rho_k$. This separation isolates two design tasks, the choice of the precoding directions $\{\bar{\mathbf{w}}_k[q]\}$ and the power allocation that ties them to the per-AP budget in (36). How both are computed depends on the level of cooperation between the APs, so the two operation modes are treated in turn: centralized operation (Section II-A1), where the CPU designs all precoders jointly and assigns one power coefficient per user, and local operation (Section II-A2), where each AP designs its own precoder and sets its powers from local information alone. In both cases the allocation depends mainly on the large-scale fading of (31) and can be held fixed over many coherence blocks, whereas the directions are recomputed every block.

Throughout this work we assume *perfect* channel state information (CSI): every precoder is built from the true channels, $\hat{\mathbf{h}}_k[q] = \mathbf{h}_k[q]$, so no channel estimation is performed and there is no pilot contamination. The uplink-pilot and

TDD-reciprocity mechanism described below is therefore the practical way this CSI would be acquired, not a source of estimation error in the present model; modelling pilot-based estimation and its error covariance is left to future work. The pilot overhead is still charged to the rate, through the frame parameters of Section I-B: the pilots occupy the uplink signalling phase, of duration $\tau_{\text{UL}} \tau_{\text{UL},\text{sig}}$ relative to the frame, and the downlink rate (28) carries the prelog $\tau_{\text{DL}}(1 - \tau_{\text{DL},\text{sig}})$. The downlink-only evaluation used here, in which the uplink phase carries nothing but pilots and the downlink phase nothing but data, is the case $\tau_{\text{UL},\text{sig}} = 1$ and $\tau_{\text{DL},\text{sig}} = 0$, so that the pilots take up a fraction τ_{UL} of the frame and the downlink prelog is $\tau_{\text{DL}} = 1 - \tau_{\text{UL}}$.

1) *Centralized Operation:* In centralized operation the APs forward their received uplink pilots to the CPU, which estimates the collective channels $\{\mathbf{h}_k[q]\}$ and designs all precoders jointly [4]. Exploiting TDD reciprocity, these uplink estimates serve as the downlink CSI, so no downlink pilots are needed. Because the CPU sees the whole LM -antenna array, it can make the serving APs cancel one another's interference, using up to LM spatial degrees of freedom per user. Collecting the user channels into $\mathbf{H}[q] = [\mathbf{h}_1[q] \cdots \mathbf{h}_K[q]] \in \mathbb{C}^{LM \times K}$, the zero forcing (ZF) precoder picks the directions that null all inter-user interference,

$$\bar{\mathbf{W}}[q] = \mathbf{H}[q](\mathbf{H}[q]^H \mathbf{H}[q])^{-1} \in \mathbb{C}^{LM \times K}, \quad (39)$$

whose column $\bar{\mathbf{w}}_k[q]$ is orthogonal to every other user's channel when the CSI is accurate; this requires $K \leq LM$ so that the Gram matrix $\mathbf{H}[q]^H \mathbf{H}[q]$ is invertible. Zero-forcing disregards the noise, so at low SNR it is improved by adding a diagonal loading λ to the Gram matrix, giving the regularized family

$$\bar{\mathbf{W}}[q] = \mathbf{H}[q](\mathbf{H}[q]^H \mathbf{H}[q] + \lambda \mathbf{I}_K)^{-1}, \quad (40)$$

which trades a little residual interference for robustness. With the loading set to the noise power, $\lambda = \sigma^2$, this is the downlink minimum mean squared error (MMSE) precoder, the dual of MMSE combining; under the perfect-CSI assumption used here the MMSE precoder coincides with regularized zero forcing (RZF) at that loading, and the two differ only once a channel-estimation-error covariance is modelled. Zero-forcing, the $\lambda = 0$ limit, is the scheme adopted in the power model of Section I.

Because a centralized precoder is designed jointly across the APs, a single power coefficient $\rho_k \geq 0$ is attached to each user and is common to all APs; the direction $\bar{\mathbf{w}}_k[q]$ already fixes how that power is split over the LM antennas, and the direction is scaled to radiate ρ_k as in (38). Two nominal allocations set $\{\rho_k\}$, both spending the total network budget $P_{\text{tot}} = L\rho_{\max}$. Equal power gives every user the same share, $\rho_k = P_{\text{tot}}/K$, whereas fractional power control weights users by their aggregate large-scale gain $\beta_k = \sum_{l=1}^L \beta_{kl}$,

$$\rho_k = P_{\text{tot}} \frac{\beta_k^v}{\sum_{i=1}^K \beta_i^v}, \quad (41)$$

with the exponent v trading throughput ($v > 0$, favouring strong users) against fairness ($v < 0$, protecting weak users).

These nominal powers make the per-AP loads average to $\rho_{\max}$ but do not by themselves respect each per-AP constraint in (36). Since the joint directions cannot be rescaled per AP without breaking the interference nulling, a single common factor

$$\gamma = \sqrt{\frac{\rho_{\max}}{\max_l \tilde{p}_l}}, \quad \tilde{p}_l = \sum_{q=1}^Q \sum_{k=1}^K \mathbb{E} \left(\|\mathbf{w}_{kl}[q]\|^2 \right), \quad (42)$$

rescales all precoders, where $\tilde{p}_l$ is the power AP l would radiate under the nominal allocation. The busiest AP then transmits at exactly $\rho_{\max}$ and the others stay below their budget, while the common scalar preserves the relative amplitudes and phases across APs and hence the interference nulling of (39). Because $\sum_k \rho_k = L\rho_{\max}$, the per-AP powers average to $\rho_{\max}$, so $\gamma \leq 1$ and the constraint is always feasible.

2) *Local Operation*: In local (distributed) operation the CPU only encodes the data, and each AP forms its own precoding directions from its locally estimated channels $\hat{\mathbf{h}}_{kl}[q]$, without sharing instantaneous CSI. An AP can then suppress only the interference it itself generates, using the M spatial degrees of freedom of its own array, so the coherent cross-AP cancellation of the centralized case is lost. The simplest choice is maximum ratio (MR) transmission, which points each beam along the local channel,

$$\bar{\mathbf{w}}_{kl}[q] = \mathbf{h}_{kl}[q], \quad (43)$$

maximizing the signal power delivered to the desired user but ignoring the interference it causes. Interference is instead suppressed locally by the counterpart of zero-forcing applied to each AP's own users. Writing $\mathbf{E}_l[q] = [\mathbf{h}_{1l}[q] \cdots \mathbf{h}_{Kl}[q]] \in \mathbb{C}^{M \times K}$ for AP l 's local channel matrix, the per-AP regularized precoder is

$$\mathbf{W}_l[q] = (\mathbf{E}_l[q]\mathbf{E}_l[q]^H + \lambda \mathbf{I}_M)^{-1} \mathbf{E}_l[q] \in \mathbb{C}^{M \times K}, \quad (44)$$

$$l = 1, \dots, L, \quad (45)$$

whose column k is the local direction $\bar{\mathbf{w}}_{kl}[q]$, and the collective direction $\bar{\mathbf{w}}_k[q]$ stacks the L per-AP blocks. Local zero-forcing is the $\lambda = 0$ limit, which nulls the interference an AP causes to its own users only when $M \geq K$; because each AP is small (typically $M < K$), the loading is needed to keep the $M \times M$ inverse well conditioned, giving local RZF for a heuristic λ and local MMSE for $\lambda = \sigma^2$. As in the centralized case, under perfect CSI local MMSE coincides with local RZF at that loading. The $M \times M$ antenna-domain form is used because each AP's array is small.

Because each AP designs and normalizes its own block independently, the power is parametrized per user per AP: AP l attaches a coefficient $\rho_{kl} \geq 0$ to its beam for user k and scales the block to radiate it across the OFDM block,

$$\mathbf{w}_{kl}[q] = \sqrt{\rho_{kl}} \frac{\bar{\mathbf{w}}_{kl}[q]}{\sqrt{\sum_{q'=1}^Q \mathbb{E} \left(\|\bar{\mathbf{w}}_{kl}[q']\|^2 \right)}}. \quad (46)$$

Only fractional power control is used: each AP splits its own budget $\rho_{\max}$ among its users according to their local large-scale gains,

$$\rho_{kl} = \rho_{\max} \frac{\beta_{kl}^v}{\sum_{i=1}^K \beta_{il}^v}, \quad l = 1, \dots, L, \quad (47)$$

so that $\sum_{k=1}^K \rho_{kl} = \rho_{\max}$ by construction and every AP meets its per-array budget in (36) with equality, using only local quantities and without any CPU-side global rescaling. The exponent v plays the same throughput-versus-fairness role as above, with $v = 1/2$ recovering the heuristic of [4]. Beyond such statistics-based rules the powers could be optimized for a network utility such as max-min fairness or sum rate, but those problems do not scale to very large networks, so the heuristics above are the practical default.

B. Complexity of precoders

TODO compute complexity of the precoders so this can be used in the power model

III. DISTRIBUTED MASSIVE MIMO UPLINK SYSTEM MODEL

We now consider the uplink of the same distributed (cell-free) massive MIMO network as in Section II. The K users now transmit simultaneously on the same time-frequency resources, each AP observes a superposition of all of them on its M antennas, and the per-AP observations are combined into one soft estimate per user. Under TDD operation the two directions occupy the same band within one coherence block, so the propagation channels are the same $\mathbf{h}_{kl}[q]$ as in (30) and the large-scale fading coefficients β_{kl} of (31) are unchanged. The downlink model applies the array response as $\mathbf{h}_k[q]^H$, so reciprocity makes $\mathbf{h}_k[q]$ itself the channel that carries the signal of user k to the LM distributed antennas.

User k transmits $x_k[q] = \sqrt{p_k[q]} s_k[q]$ on subcarrier q , where $s_k[q] \in \mathbb{C}$ is its unit-power data symbol, $\mathbb{E}(|s_k[q]|^2) = 1$, mutually independent across users, and $p_k[q] \geq 0$ is the power it spends on that subcarrier. Each user is its own transmitter and is subject to its own budget over the OFDM block,

$$\sum_{q=1}^Q p_k[q] = p_k \leq p_{\max}, \quad k = 1, \dots, K, \quad (48)$$

with $p_{\max}$ the maximum transmit power of a user terminal. Since the power control below acts on the large-scale fading, which is flat over the band, the power is spread uniformly over the subcarriers, $p_k[q] = p_k/Q$.

The signal received by AP l on subcarrier q is

$$\mathbf{y}_l[q] = \sum_{k=1}^K \sqrt{p_k[q]} \mathbf{h}_{kl}[q] s_k[q] + \mathbf{n}_l[q] \in \mathbb{C}^M, \quad (49)$$

where $\mathbf{n}_l[q] \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I}_M)$ is the receiver noise, independent across APs and subcarriers and independent of the channels and data. The noise power σ^2 is the same quantity

as in (34), but it is now referred to the AP receive chain, so it is the one set by the noise figure of the LNAs and analog front end of Section I-D. Stacking the L observations gives the collective received vector

$$\mathbf{y}[q] = \begin{bmatrix} \mathbf{y}_1[q] \\ \vdots \\ \mathbf{y}_L[q] \end{bmatrix} = \mathbf{H}[q] \mathbf{P}[q]^{1/2} \mathbf{s}[q] + \mathbf{n}[q] \in \mathbb{C}^{LM}, \quad (50)$$

with $\mathbf{H}[q] = [\mathbf{h}_1[q] \cdots \mathbf{h}_K[q]] \in \mathbb{C}^{LM \times K}$ the collective channel matrix already used in Section II-A1, $\mathbf{s}[q] = [s_1[q] \cdots s_K[q]]^\top \in \mathbb{C}^K$, $\mathbf{n}[q] \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I}_{LM})$, and $\mathbf{P}[q]^{1/2} \in \mathbb{C}^{K \times K}$ the diagonal matrix with $[\mathbf{P}[q]^{1/2}]_{k,k} = \sqrt{p_k[q]}$ and zero off-diagonal entries.

The data symbol of user k is estimated by applying a receive combining vector $\mathbf{v}_k[q] \in \mathbb{C}^{LM}$ to (50), which separates into the desired term, the multi-user interference, and the noise,

$$\begin{aligned} \hat{s}_k[q] &= \mathbf{v}_k[q]^H \mathbf{y}[q] \\ &= \underbrace{\sqrt{p_k[q]} \mathbf{v}_k[q]^H \mathbf{h}_k[q] s_k[q]}_{\text{desired signal}} \\ &\quad + \underbrace{\sum_{\substack{i=1 \\ i \neq k}}^K \sqrt{p_i[q]} \mathbf{v}_k[q]^H \mathbf{h}_i[q] s_i[q]}_{\text{multi-user interference}} + \underbrace{\mathbf{v}_k[q]^H \mathbf{n}[q]}_{\text{noise}}. \end{aligned} \quad (51)$$

Writing $\mathbf{v}_k[q] = [\mathbf{v}_{k1}[q]^\top \cdots \mathbf{v}_{kL}[q]^\top]^\top$ in blocks of M entries, with $\mathbf{v}_{kl}[q] \in \mathbb{C}^M$ the part applied to AP l , the estimate is the sum of per-AP contributions,

$$\hat{s}_k[q] = \sum_{l=1}^L \mathbf{v}_{kl}[q]^H \mathbf{y}_l[q]. \quad (52)$$

Linear combining is therefore distributive over the APs, and this is the property the functional splits of Section IV-E exploit: an AP that holds its own combining coefficients can form its partial sum locally and forward one scalar stream per user instead of M sample streams.

Treating the multi-user interference in (51) as noise, and using that the noise term has variance $\sigma^2 \|\mathbf{v}_k[q]\|^2$, the effective uplink SINR of user k on subcarrier q is

$$\text{SINR}_k^{\text{ul}}[q] = \frac{p_k[q] |\mathbf{v}_k[q]^H \mathbf{h}_k[q]|^2}{\sum_{\substack{i=1 \\ i \neq k}}^K p_i[q] |\mathbf{v}_k[q]^H \mathbf{h}_i[q]|^2 + \sigma^2 \|\mathbf{v}_k[q]\|^2}. \quad (53)$$

With the effective channel known at the decoder, an achievable rate for user k on subcarrier q is $\log_2(1 + \text{SINR}_k^{\text{ul}}[q])$, and inserting these per-user, per-subcarrier SINRs into the ergodic rate expression of Section I-F ((28)) gives the delivered per-user uplink rate

$$\begin{aligned} R_{\text{UL},k} &= \bar{x}_{\text{UL}} \tau_{\text{UL}} (1 - \tau_{\text{UL},\text{sig}}) \frac{\bar{B}}{\bar{Q}} \\ &\quad \times \mathbb{E} \left(\sum_{q=1}^Q \log_2(1 + \text{SINR}_k^{\text{ul}}[q]) \right), \end{aligned} \quad (54)$$

with $R_{\text{UL}} = \sum_{k=1}^K R_{\text{UL},k}$ the delivered uplink sum rate. It drives the channel-decoder consumption of Section I-E and, through (88), the traffic-dependent part of the fronthaul.

A. Combiner and Power Control Designs

As in the downlink, the design splits into the choice of the receive directions $\{\mathbf{v}_k[q]\}$, which depends on how much the APs cooperate, and the choice of the transmit powers $\{p_k\}$, which is a per-user allocation driven by the large-scale fading. The two are treated in turn, and the directions are recomputed every coherence block whereas the powers can be held fixed over many blocks.

Throughout, and consistently with Section II-A, we assume *perfect* CSI: every combiner is built from the true channels, $\hat{\mathbf{h}}_k[q] = \mathbf{h}_k[q]$, so there is no estimation error and no pilot contamination. The pilot overhead is nevertheless charged to the rate through the frame parameters of Section I-B. In the downlink-only evaluation of Section II-A the uplink phase carried nothing but pilots, $\tau_{\text{UL},\text{sig}} = 1$, which leaves (54) at zero. An uplink that carries data requires $\tau_{\text{UL},\text{sig}} < 1$, and the coherence block then has to be split explicitly: with τ_c samples per block of which τ_p are pilots, the signalling fraction of the frame is $\tau_{\text{UL}} \tau_{\text{UL},\text{sig}} = \tau_p / \tau_c$ and the uplink data prelog of (54) is $\tau_{\text{UL}}(1 - \tau_{\text{UL},\text{sig}}) = \tau_{\text{UL}} - \tau_p / \tau_c$, which requires $\tau_p / \tau_c \leq \tau_{\text{UL}}$. Mutually orthogonal pilots additionally require $\tau_p \geq K$; this is what makes the no-contamination assumption above self-consistent, and it is the assumption under which the perfect-CSI model is the idealisation of a real pilot phase rather than a different system.

1) *Centralized Operation:* In centralized operation the APs forward their received samples to the central processing unit (CPU), which holds $\mathbf{y}[q]$ in full and combines over all LM antennas jointly [4]. This is the highest level of cooperation, the receive counterpart of the joint precoding of Section II-A1, and it corresponds to splits S1 and S2 of Section IV-E. Minimizing the mean squared error $\mathbb{E}(|s_k[q] - \hat{s}_k[q]|^2)$ over $\mathbf{v}_k[q]$ for the model (50) gives the MMSE combiner

$$\mathbf{v}_k[q] = (\mathbf{H}[q] \mathbf{P}[q] \mathbf{H}[q]^H + \sigma^2 \mathbf{I}_{LM})^{-1} \mathbf{h}_k[q], \quad (55)$$

which requires inverting an $LM \times LM$ matrix. Applying the push-through identity $(\mathbf{A}\mathbf{B} + \sigma^2 \mathbf{I})^{-1} \mathbf{A} = \mathbf{A}(\mathbf{B}\mathbf{A} + \sigma^2 \mathbf{I})^{-1}$ with $\mathbf{A} = \mathbf{H}[q] \mathbf{P}[q]$ and $\mathbf{B} = \mathbf{H}[q]^H$, and discarding the diagonal factor that the scale invariance of (53) makes irrelevant, the same directions are obtained from a $K \times K$ inverse,

$$\mathbf{V}[q] = \mathbf{H}[q] (\mathbf{H}[q]^H \mathbf{H}[q] + \sigma^2 \mathbf{P}[q]^{-1})^{-1} \in \mathbb{C}^{LM \times K}, \quad (56)$$

whose column k is $\mathbf{v}_k[q]$ up to a scalar. This is the uplink form of the regularized family (40), with the heuristic loading $\lambda \mathbf{I}_K$ replaced by the physically determined $\sigma^2 \mathbf{P}[q]^{-1}$; for equal powers $p_k[q] = p$ the two coincide at $\lambda = \sigma^2 / p$, which is the sense in which the downlink RZF precoder is the dual of MMSE combining. Letting the loading vanish, which is the high-SNR limit $\sigma^2 / p \rightarrow 0$, gives the ZF combiner

$$\mathbf{V}[q] = \mathbf{H}[q] (\mathbf{H}[q]^H \mathbf{H}[q])^{-1}, \quad (57)$$

for which $\mathbf{v}_k[q]^H \mathbf{h}_i[q] = 0$ for all $i \neq k$, so all inter-user interference is removed and (53) reduces to $p_k[q]/(\sigma^2 \|\mathbf{v}_k[q]\|^2)$. As in the downlink this needs $K \leq LM$ for the Gram matrix to be invertible, and it disregards the noise, so it is the scheme of choice only when the loading is negligible. Zero-forcing is the scheme adopted in the power model of Section I and its distributed extension.

Because the combiner is scale-invariant, the centralized design carries no power allocation of its own; the transmit powers are set at the users by (63) below. The CPU does need the powers $\{p_k\}$ to build (56), but these are large-scale quantities refreshed once per large-scale fading realisation, not once per coherence block.

2) *Local Operation and Large-Scale Fading Decoding*: In local operation each AP combines its own observation with its own CSI and forwards one scalar per user, so no instantaneous CSI crosses the fronthaul. This is split S3 of Section IV-E. AP l forms $\hat{s}_{kl}[q] = \mathbf{v}_{kl}[q]^H \mathbf{y}_l[q]$ using either MR combining, which points along the local channel,

$$\mathbf{v}_{kl}[q] = \mathbf{h}_{kl}[q], \quad (58)$$

or the local regularized combiner built from AP l 's own channel matrix $\mathbf{E}_l[q] = [\mathbf{h}_{1l}[q] \cdots \mathbf{h}_{Kl}[q]] \in \mathbb{C}^{M \times K}$,

$$\mathbf{V}_l[q] = (\mathbf{E}_l[q] \mathbf{P}[q] \mathbf{E}_l[q]^H + \sigma^2 \mathbf{I}_M)^{-1} \mathbf{E}_l[q] \in \mathbb{C}^{M \times K}, \quad (59)$$

$$l = 1, \dots, L, \quad (60)$$

whose column k is $\mathbf{v}_{kl}[q]$. This is (44) with the heuristic loading again replaced by its physical value, and it inverts an $M \times M$ matrix because the per-AP array is small. An AP can suppress only the interference it observes itself, using the M degrees of freedom of its own array, so the coherent cross-AP interference rejection of (56) is lost; the loading is what keeps the inverse well conditioned in the usual regime $M < K$.

What remains is how the CPU fuses the L local estimates. It applies one weight per AP and user,

$$\hat{s}_k[q] = \sum_{l=1}^L a_{kl}[q]^* \hat{s}_{kl}[q], \quad (61)$$

collected into $\mathbf{a}_k[q] = [a_{k1}[q] \cdots a_{kL}[q]]^T \in \mathbb{C}^L$. By (52) this is exactly the collective combiner $\mathbf{v}_k[q]$ whose l -th block is $a_{kl}[q] \mathbf{v}_{kl}[q]$, so (53) continues to apply unchanged and the fusion rule is part of the combiner rather than a separate stage. Two choices are used. Equal weighting, $a_{kl}[q] = 1$, simply adds the local estimates and needs nothing at the CPU beyond the streams themselves. Large-scale fading decoding (LSFD) instead optimizes the weights from the channel statistics: collecting the per-AP effective gains $g_{kil}[q] \triangleq \mathbf{v}_{kl}[q]^H \mathbf{h}_{il}[q]$ into $\mathbf{g}_{ki}[q] = [g_{ki1}[q] \cdots g_{kiL}[q]]^T \in \mathbb{C}^L$, and writing $\mathbf{D}_k[q]$ for the diagonal matrix with entries $\|\mathbf{v}_{kl}[q]\|^2$ (diagonal because the noise is independent across APs), the weights that maximize the resulting use-and-then-forget SINR are

$$\mathbf{a}_k[q] = \left(\sum_{i=1}^K p_i[q] \mathbb{E}(\mathbf{g}_{ki}[q] \mathbf{g}_{ki}[q]^H) + \sigma^2 \mathbb{E}(\mathbf{D}_k[q]) \right)^{-1} \times \sqrt{p_k[q]} \mathbb{E}(\mathbf{g}_{kk}[q]), \quad (62)$$

where the expectations are over the small-scale fading for fixed user positions [4]. **TODO verify (62) against the LSFD expression in [4] before it is used in the evaluation.** These weights are statistical, not instantaneous, so they are computed once per large-scale fading realisation and cost the fronthaul nothing per coherence block. The three fusion rules therefore span three cooperation levels at the same per-AP combiner: local combining at a single AP, equal-weight summation at the CPU, and statistically weighted summation. Only the centralized combiner of (56) uses instantaneous CSI at the CPU, and the SE expressions of the different levels must not be mixed.

Two things should be flagged about how (53) is used at these lower levels. When the fusion weights are statistical, the CPU does not know the instantaneous effective channel $\mathbf{v}_k[q]^H \mathbf{h}_k[q]$, and the rigorous achievable rate is then the use-and-then-forget bound, in which the mean effective channel is treated as the useful gain and its fluctuation as extra interference. Evaluating (54) with the instantaneous (53), as is done in the downlink of Section II, assumes instead that the effective channel is known wherever the decoding happens, and is therefore optimistic for local operation by the amount of residual channel hardening. **TODO decide whether the local-operation rate is reported with the instantaneous SINR, for consistency with the downlink, or with the use-and-then-forget bound, which is the achievable one; the gap grows as M shrinks.**

3) *Uplink Power Control*: The powers $\{p_k\}$ are set at the users from the large-scale fading and, by (48), are constrained one user at a time. Full power, $p_k = p_{\max}$, is the natural reference, but it is the worst case for fairness: a user close to an AP then swamps a distant one sharing the same resources. Fractional power control weights the users by their aggregate large-scale gain $\beta_k = \sum_{l=1}^L \beta_{kl}$,

$$p_k = p_{\max} \frac{\beta_k^v}{\max_i \beta_i^v}, \quad k = 1, \dots, K, \quad (63)$$

which satisfies $p_k \leq p_{\max}$ for either sign of the exponent v , since the normalizing maximum is attained by the strongest user when $v > 0$ and by the weakest when $v < 0$. The exponent plays the same throughput-versus-fairness role as in (41), with $v = 0$ recovering full power and $v = -1$ inverting the channel statistically, so that every user arrives at the network with the same average total gain $p_k \beta_k$. The normalization by a maximum rather than by a sum is what differs from the downlink rule (41): there a shared budget had to be divided among the users, whereas here each user has its own budget and the only role of the denominator is to keep the strongest user within it. As a result the uplink has no counterpart of the common rescaling (42), and no feasibility question arises.

The uplink transmit power is spent at the users and therefore does not appear in the network consumption P_{net} of (64), which counts the APs, the fronthaul, and the CPU only. The asymmetry this creates in the power model is worth stating plainly: in the downlink phase the APs pay for the PAs of

Section IV-C, whose consumption scales with the radiated power, whereas in the uplink phase they pay only for the receive front end (75) and the digital processing (77), both of which are essentially independent of $p_{\max}$. Uplink power control therefore acts on the delivered rate (54) and on nothing else in (94), which makes it a pure spectral-efficiency lever rather than an energy one, and it is the reason the uplink and downlink phases of the same frame have very different consumption profiles.

B. Complexity of combiners

TODO compute complexity of the combiners so this can be used in the power model; under TDD reciprocity the centralized combiner reuses the matrix already inverted for (79), so only the application cost $2K'N$ should be charged in the uplink, whereas LSFD adds the per-user $L \times L$ solve of (62) once per large-scale fading realisation.

IV. D-MIMO POWER MODEL

The model of Section I describes one central site: a single co-located array whose antennas, converters, and processors share a physical location, a clock, and a power supply. The distributed system of Section II replaces that site by L APs of M antennas each, spread over the coverage area and tied to a CPU by fronthaul links. Three things change as a result. First, the hardware blocks become per-AP quantities that must be summed over the network, each at its own operating point. Second, the transport between the APs and the CPU has its own power consumption, which is not modeled in the single-site model. Third, the digital processing chain is split across two locations (AP and CPU). This requires design choices about where a given operation runs. This section extends the parametric model along those three axes while keeping the frame-averaging machinery of Section I-B intact, so that the single-site model is recovered as a special case by setting $L = 1$ with no fronthaul and no CPU.

Throughout, $l = 1, \dots, L$ indexes APs, $k = 1, \dots, K$ users, $q = 1, \dots, Q$ subcarriers, and $i \in \{\text{DL}, \text{UL}\}$ the link direction, matching Section II. Hardware parameters carried over from Section I keep their symbols and now refer to a single AP. The number of data subcarriers is written q_i in (28) and $Q_{\max}$ in (2); we use Q for it throughout and reserve q for the subcarrier index. TODO align the subcarrier-count symbol across Section I and Section II.

A. Network power decomposition

Let $\mathcal{S} \subseteq \{1, \dots, L\}$ be the set of APs that are switched on; an AP outside $\mathcal{S}$ is in deep sleep and draws a fraction $\delta^{\text{ds}} \in [0, 1]$ of its base consumption $P_{\text{AP},l}^0$. The total distributed network consumption is

$$P_{\text{net}} = \sum_{l \in \mathcal{S}} \left(P_{\text{AP},l} + \frac{\overline{P_{\text{FH},l}}}{\eta_{\text{FH},\text{sc}}} \right) + \sum_{l \notin \mathcal{S}} \delta^{\text{ds}} \left(P_{\text{AP},l}^0 + \frac{P_{\text{FH},l}^0}{\eta_{\text{FH},\text{sc}}} \right) + \frac{\overline{P_{\text{CPU}}}}{\eta_{\text{CPU},\text{sc}}}, \quad (64)$$

where the consumption of an active AP retains the three-block structure of (1),

$$P_{\text{AP},l} = \frac{\overline{P_{\text{dig},l}}}{\eta_{\text{dig},\text{sc}}} + \frac{\overline{P_{\text{ana},l}}}{\eta_{\text{ana},\text{sc}}} + \frac{\overline{P_{\text{PA},l}}}{\eta_{\text{PA},\text{sc}}}, \quad (65)$$

and $\eta_{\text{FH},\text{sc}}, \eta_{\text{CPU},\text{sc}} \in (0, 1]$ are the supply-and-cooling efficiencies of the fronthaul transceivers and of the central unit. The CPU efficiency plays the role of an inverse power-usage effectiveness: the central unit sits in a room with its own cooling, which is not the same environment as an outdoor AP enclosure, so $\eta_{\text{CPU},\text{sc}}$ should not be set equal to the $\eta_{\text{c},\text{sc}}$ values fitted for a macro base station. TODO find proper values, look at power usage effectiveness (PUE) of datacenters Setting $L = 1$, $\mathcal{S} = \{1\}$, $M = M_{\text{ant}}$, and dropping the fronthaul and CPU terms returns (1) unchanged.

The set $\mathcal{S}$ determines how many APs serve the network. As a first step we assume that all L APs are active.

TODO see if we want to keep the next paragraph: The set $\mathcal{S}$ is the first genuinely new degree of freedom. A co-located array can shut down individual chains but not the site; a distributed deployment can shut down whole APs while the network keeps serving, and the coverage redundancy of cell-free operation is what makes that possible. AP selection therefore acts directly on (64), and Remark 6 shows that it also relieves the fronthaul.

B. Serving clusters, per-AP load, and the shared frame

In the fully distributed operation of Section II every AP serves every user. The power model should not assume this, because restricting each AP to a subset of users is the standard way of keeping cell-free operation scalable and is also the main lever on fronthaul traffic. Let $\mathcal{D}_l \subseteq \{1, \dots, K\}$ be the set of users served by AP l and $\mathcal{L}_k = \{l : k \in \mathcal{D}_l\}$ the serving cluster of user k , with $K_l \triangleq |\mathcal{D}_l|$ and $|\mathcal{L}_k|$ its cardinality. Full cell-free operation is the special case $\mathcal{D}_l = \{1, \dots, K\}$ for all l , so $K_l = K$ and $|\mathcal{L}_k| = L$. A user outside $\mathcal{D}_l$ has $\mathbf{w}_{kl}[q] = \mathbf{0}$, which leaves the signal model of (32) untouched.

The frame timing, in contrast, is shared. Under TDD the APs must switch between directions together, since an AP still transmitting while a neighbour receives creates cross-link interference between nodes that are far closer to each other than to any user. The frame parameters $\tau_{\text{DL}}, \tau_{\text{UL}}, \tau_{i,\text{sig}}$ of Section I-B are therefore network-wide constants, and only the load is allowed to differ between APs. Repeating the construction of (3) over the users AP l actually serves gives the per-AP average physical resource load

$$\bar{x}_{i,l} = \frac{N_{a,i}}{N_i} \frac{K_{i,l}}{K_{\max}}, \quad (66)$$

with $K_{i,l} = |\mathcal{D}_l|$ counted over the users active in direction i and $K_{\max}$ the network-wide number of spatial streams. The frame-averaging function (6) is reused with this per AP load,

$$\begin{aligned} \mathcal{A}_{i,l}^c(P^a, P^s, P^0) &\triangleq \bar{x}_{i,l} \tau_i (1 - \tau_{i,\text{sig}}) P^a + \tau_i \tau_{i,\text{sig}} P^s \\ &\quad + (1 - \bar{x}_{i,l}) \tau_i (1 - \tau_{i,\text{sig}}) \delta_c^\mu P^0 \\ &\quad + (1 - \tau_i) \delta_c^0 P^0, \end{aligned} \quad (67)$$

and the shorthand $\mathcal{A}_{i,l}^c(P) \triangleq \mathcal{A}_{i,l}^c(P, P, P)$ is kept for the blocks whose consumption does not depend on the operating mode. Under full cell-free operation all APs share the same load, $\bar{x}_{i,l} = \bar{x}_i$, and (67) collapses to (6).

C. Per-AP power amplifiers

Each AP is an independent transmitter with its own budget (36), so the single quantity $P_{T,DL}$ of Section I-C becomes a vector of per-AP transmit powers. From the precoders of Section II-A, the power actually radiated by AP l over the OFDM block is

$$P_{T,l} \triangleq \sum_{q=1}^Q \sum_{k=1}^K \mathbb{E} \left(\|\mathbf{w}_{kl}[q]\|^2 \right) \leq \rho_{\max}, \quad (68)$$

which the two operation modes fill in differently. Local operation meets the constraint with equality by construction, $P_{T,l} = \rho_{\max}$ for every AP, because (47) splits each AP's own budget among its users. Centralized operation cannot rescale APs independently without breaking the interference nulling, so the common factor of (42) gives $P_{T,l} = \gamma^2 \tilde{p}_l$ with $\gamma^2 = \rho_{\max} / \max_l \tilde{p}_l$, and only the busiest AP reaches its budget. It is convenient to work with the per-AP power utilisation

$$u_l \triangleq \frac{P_{T,l}}{\rho_{\max}} \in [0, 1]. \quad (69)$$

As was done in the centralized power model we assume that the per-AP power is shared equally across its antennas, the per-PA output power is $P_{a,l} = P_{T,l}/M$ and the frame-averaged PA consumption of AP l follows (8) with the per-AP load,

$$\overline{P_{PA,l}} = M \mathcal{A}_{DL,l}^{PA}(P_{PA}(P_{a,l}), P_{PA}(\zeta_{\text{sig}} P_{\max}), P_{PA}(0)), \quad (70)$$

with $P_{\max}$ the maximum PA output power and ζ_{sig} the fraction of it used during signalling, both as in (8).

Note that the equal-split assumption is not exact: the per-antenna powers $\sum_q \sum_k \mathbb{E} \left(|[\mathbf{w}_{kl}[q]]_m|^2 \right)$ differ across $m = 1, \dots, M$ for any precoder that is not isotropic. Because the efficiency exponent $\alpha \in [0, 1]$ of (7) makes $p \mapsto p^\alpha$ concave, and hence $P_{PA}(\cdot)$ concave, Jensen's inequality gives $\frac{1}{M} \sum_m P_{PA}(P_{a,lm}) \leq P_{PA}(P_{a,l})$: the equal-split assumption is a conservative upper bound on the true PA consumption, and it is tight when the precoder loads the antennas of an AP uniformly.

Remark 2 (PA sizing and the static floor). *The relation between $P_{\max}$ and the AP budget must be stated, because it decides how the PA static term scales with the deployment. If each PA is dimensioned for its share of the AP budget, $P_{\max} = \rho_{\max}/M$, then substituting (7) into (70) makes the data-mode PA consumption of AP l independent of M ,*

$$M P_{PA}(P_{a,l}) = \frac{\rho_{\max}}{\eta_{PA,\max}} \left[\xi + (1 - \xi) u_l^\alpha \right], \quad (71)$$

where $\xi \in [0, 1]$ is the static-to-dynamic consumption ratio and $\eta_{PA,\max}$ the maximum PA efficiency of (7). A fully loaded AP therefore consumes exactly $\rho_{\max}/\eta_{PA,\max}$, and the network static floor is $L \xi \rho_{\max}/\eta_{PA,\max}$, growing with the number of

APs rather than with the number of antennas. If instead $P_{\max}$ is a fixed component rating independent of M , the floor is $LM \xi P_{\max}/\eta_{PA,\max}$ and grows with the total antenna count. The two conventions differ by a factor $MP_{\max}/\rho_{\max}$ and must not be mixed. The three PA parameters ξ , α , and $\eta_{PA,\max}$ were fitted for macro-cell amplifiers and should be refitted for the small amplifiers of a distributed AP rather than reused. **TODO** source $\xi, \alpha, \eta_{PA,\max}$ for AP-class PAs.

Remark 3 (Cost of the per-AP constraint). **TODO check this** Under centralized operation with either nominal allocation of Section II-A1, the nominal powers satisfy $\sum_l \tilde{p}_l = \sum_k \rho_k = L\rho_{\max}$, so the average utilisation is

$$\bar{u} \triangleq \frac{1}{L} \sum_{l=1}^L u_l = \frac{\gamma^2}{L\rho_{\max}} \sum_{l=1}^L \tilde{p}_l = \gamma^2. \quad (72)$$

The common rescaling factor of (42) is therefore exactly the fraction of the network power budget that is actually radiated. Combining (71) with (72) and the concavity of $u \mapsto u^\alpha$ bounds the network PA consumption in the data mode from below by

$$\sum_{l=1}^L M P_{PA}(P_{a,l}) \geq \frac{L\rho_{\max}}{\eta_{PA,\max}} \left[\xi + (1 - \xi) \gamma^{2\alpha} \right], \quad (73)$$

with equality when all APs are equally loaded. The per-AP constraint thus costs twice: the network radiates only a fraction γ^2 of $L\rho_{\max}$, yet the static floor $L\xi\rho_{\max}/\eta_{PA,\max}$ is paid in full whenever the PAs are not asleep. An imbalanced deployment, in which one AP is much busier than the rest and drives γ down, is penalised on both counts.

D. Per-AP analog front end

A distributed AP carries few antennas, so hybrid beam-forming has nothing to amortise and the arrays are fully digital: $M_{\text{RF}} = M$ and $M_{\text{PS}} = 0$. Specialising (9) and (10) accordingly, the working-mode analog consumption of one AP is

$$P_{\text{ana},DL,l} = P_{\text{LO}} + M(2P_{\text{DAC}} + 2P_{\text{filt},\text{RF}} + 2P_{\text{mix}}), \quad (74)$$

$$P_{\text{ana},UL,l} = P_{\text{LO}} + M(2P_{\text{ADC}} + 2P_{\text{filt},\text{RF}} + 2P_{\text{mix}} + P_{\text{LNA}}), \quad (75)$$

and the frame-averaged analog block of AP l is

$$\begin{aligned} \overline{P_{\text{ana},l}} &= \mathcal{A}_{DL,l}^{\text{ana}}(P_{\text{ana},DL,l}) + \mathcal{A}_{UL,l}^{\text{ana}}(P_{\text{ana},UL,l}) \\ &\quad + P_{\text{ana},\text{misc}} + P_{\text{sync}}. \end{aligned} \quad (76)$$

The local oscillator is the first term to change character. In the co-located model a single LO is shared by M_{ant} chains and contributes a constant; here every AP needs its own, so the network pays $L P_{\text{LO}}$ and the term grows with the deployment even though it is small per node.

The term P_{sync} has no counterpart in Section I-D and is added here. Coherent joint transmission across physically separate nodes requires the APs to share a frequency and phase reference and requires the TDD chains to be reciprocity-calibrated; the precoder of (39) nulls interference only if the relative phases across APs are the ones the CPU assumed.

TABLE I: Node at which each digital operation runs, per functional split. The uplink combiner follows the same allocation as the downlink precoder. Splits S1 and S2 both realise the centralized precoding of Section II-A1 and deliver the same rate; S3 is local operation (Section II-A2) and delivers the lower rate of local precoding. [†]Listed for completeness but not priced, since Section II assumes perfect CSI **TODO add this**.

Operation	Count	S1	S2	S3
Encoding / decoding	(17)	CPU	CPU	CPU
Channel estimation [†]	—	CPU	CPU	AP
Precoder computation	(79), (80)	CPU	CPU	AP
Precoder application	$2K'N$	CPU	AP	AP
OFDM (I)FFT	(21)	AP	AP	AP
Predistortion	(22)	AP	AP	AP
Baseband filter	(23)	AP	AP	AP

Whether the reference is distributed over the fronthaul, over a dedicated timing network, or over the air, it costs power at every AP continuously, which is why P_{sync} sits outside the direction-dependent averaging. It is a real cost of distributing the array and one that a co-located design does not pay. **TODO source a value for P_{sync} ; over-the-air calibration also consumes frame time, which is not yet charged to $\tau_{i,\text{sig}}$.**

E. Functional split of Digital Signal Processing

In the single-site model every digital operation runs at the base station. In a distributed deployment the encoder, the precoder computation, and the precoder application can each sit at the CPU or at the AP, and the choice sets both the digital consumption and the fronthaul rate. We distinguish three splits, which span the practical range and map onto the operation modes of Section II-A; Table I summarises where each operation ends up.

Split S1 forwards precoded samples. The CPU encodes, computes the precoder of (39), and applies it, sending AP l the M streams of frequency-domain samples it must radiate; the AP performs only the OFDM modulation, predistortion, baseband filtering, and conversion.

Split S2 forwards coefficients and data. The CPU encodes and computes $\mathbf{W}_l[q]$ but AP l applies it locally, so the fronthaul carries the K_l data streams plus the precoding coefficients, refreshed once per coherence block.

Split S3 is local operation as defined in Section II-A2: the CPU only encodes, and AP l estimates its own channels and computes and applies the precoder of (44) from local information, so the fronthaul carries data alone. Splits S1 and S2 both realise the centralized precoding of Section II-A1 and therefore deliver the same rate; they differ only in where the work is done and what crosses the fronthaul. Split S3 delivers the lower rate of local precoding.

F. Digital signal processing at the APs and the CPU

TODO check this The per-chain digital costs of Section I-E carry over unchanged, evaluated with $M_{\text{RF}} = M$ at each AP. Every AP performs the OFDM modulation and demodulation,

the predistortion of its own PAs, and the baseband filtering whatever the split, so

$$P_{\text{dig,DL},l} = P_{\text{IFFT}} + P_{\text{DPD}} + P_{\text{filt,BB}} + P_{\text{MIMO,DL},l}, \quad (77)$$

$$P_{\text{dig,UL},l} = P_{\text{IFFT}} + P_{\text{filt,BB}} + P_{\text{MIMO,UL},l}, \quad (78)$$

with $\overline{P}_{\text{dig},l} = \mathcal{A}_{\text{DL},l}^{\text{dig}}(P_{\text{dig,DL},l}) + \mathcal{A}_{\text{UL},l}^{\text{dig}}(P_{\text{dig,UL},l})$. Only the MIMO-processing terms $P_{\text{MIMO},i,l}$ depend on the split.

As in Section I-E, a MIMO-processing term is a static part plus $f_{s,i}\Xi/\eta_{\text{pre}}$, with Ξ its complex operations per sample. Two operations must be kept apart, because the split moves them independently. Applying an already-computed precoder to K' users over N chains costs $2K'N$ operations per sample, the count already used for the combiner in Section I-E. Computing it costs the Cholesky terms of (24), amortised over the v_{coh} samples of a coherence block. Under centralized operation the CPU inverts the $K \times K$ Gram matrix of the collective channel $\mathbf{H}[q] \in \mathbb{C}^{LM \times K}$, so (24) applies with M_{RF} replaced by the total number of distributed antennas LM ,

$$\Xi_{\text{pre}}^{\text{cen}} = \frac{1}{v_{\text{coh}}} \left(\frac{K^3}{3} + 3K^2LM + KLM \right), \quad (79)$$

written here as a network total rather than per chain. Under local operation AP l instead inverts the $M \times M$ matrix of (44), which is the same Cholesky-based accounting with the roles of the antenna and user dimensions exchanged,

$$\Xi_{\text{pre},l}^{\text{loc}} = \frac{1}{v_{\text{coh}}} \left(\frac{M^3}{3} + 3M^2K_l + MK_l \right). \quad (80)$$

Setting $L = 1$ and $M = M_{\text{RF}}$ in (79) recovers the computation part of $M_{\text{RF}}\Xi_{\text{pre}}$ in (24) exactly.

The three splits then distribute these two counts over the two nodes as

$$\Xi_i^{\text{CPU}} = \iota_{S1} 2KLM + \iota_{S1,S2} \iota_{\text{DL}} \Xi_{\text{pre}}^{\text{cen}}, \quad (81)$$

$$\Xi_{i,l}^{\text{AP}} = \iota_{S2,S3} 2K_lM + \iota_{S3} \iota_{\text{DL}} \Xi_{\text{pre},l}^{\text{loc}}, \quad (82)$$

where each $\iota_{\bullet} \in \{0,1\}$ equals one for the split or direction named in its subscript and zero otherwise, and the corresponding powers follow (19),

$$P_{\text{MIMO},i,l} = \bar{M}_{\text{RF}} P_s + \frac{f_{s,i}}{\eta_{\text{pre}}} \Xi_{i,l}^{\text{AP}}, \quad \bar{M}_{\text{RF}} = \left\lceil \frac{M}{32} \right\rceil, \quad (83)$$

and likewise for $P_{\text{MIMO},i}^{\text{CPU}}$ with Ξ_i^{CPU} . As in Section I-E, the computation is charged to the downlink only, since TDD reciprocity lets the same matrix serve the uplink combiner, while the application is charged in both directions. Every operation then appears exactly once across (81), so no work is counted twice: under S2, in particular, the AP is billed only for applying the precoder and the CPU only for computing it. Because linear combining is distributive over the APs, $\hat{s}_k = \sum_l \mathbf{v}_{kl}^H \mathbf{y}_l$, an AP holding its combining coefficients can form its own partial sum locally, so S2 attains the centralized uplink rate while forwarding K_l scalar streams instead of M sample streams.

Summed over the network, (80) grows linearly in L at fixed M while (79) grows linearly in LM and additionally carries

the K^3 term. For the dense, small- M deployments of interest the centralized cost is dominated by $3K^2LM$ and the local cost by $3M^2K_l$, so centralized precoding is more expensive by roughly a factor K/M per antenna.

Channel encoding and decoding stay at the CPU in all three splits, since that is where the payload enters and leaves the network. The CPU consumption is

$$\overline{P_{\text{CPU}}} = P_{\text{CPU},0} + \mathcal{A}_{\text{DL}}^{\text{dig}}(P_{\text{enc}} + P_{\text{MIMO,DL}}^{\text{CPU}}) + \mathcal{A}_{\text{UL}}^{\text{dig}}(P_{\text{dec}} + P_{\text{MIMO,UL}}^{\text{CPU}}), \quad (84)$$

with P_{enc} and P_{dec} from (17) driven by the network sum rates R_{DL} and R_{UL} , and $P_{\text{CPU},0}$ the always-on consumption of the central unit. The averaging in (84) uses the network-wide load $\bar{x}_i$, not a per-AP one.

Remark 4 (Loss of static-power amortisation). *The precoder, combiner, and baseband filter of Section I-E share a static power P_s per FPGA, one per $M_{\text{RF}} = \lceil M_{\text{RF}}/32 \rceil$ group of chains. A co-located array of M_{ant} antennas amortises that static power over 32 chains at a time; a distributed network of L APs with $M < 32$ antennas each pays it L times, once per AP, for a total of LM chains. At $L = 40$ and $M = 4$ the network runs 40 FPGAs where a co-located array of the same 160 antennas would run 5, an eightfold increase in this static term. Chopping a large array into many small ones leaves the per-chain digital costs untouched but destroys the sharing of the fixed costs, and this is a general feature rather than an artefact of the value 32.*

G. Fronthaul consumption

Following the cell-free power model of [5], the fronthaul link serving AP l consumes a traffic-independent part and a part proportional to the rate it carries,

$$P_{\text{FH},l} = P_{\text{FH},0} + \Pi_{\text{FH}} R_{\text{FH},l}, \quad (85)$$

with $P_{\text{FH},0}$ in watts, the traffic-dependent coefficient Π_{FH} in W per bit/s, and $R_{\text{FH},l}$ the fronthaul rate in bit/s. The static part covers the optical transceivers at both ends and depends on the link length and the topology; the traffic part covers the throughput-dependent consumption of the transport chain. In the parametrisation of [5] these are $P_{\text{FH},0} = 0.825$ W per link and $\Pi_{\text{FH}} = 0.25$ W per Gbit/s. **TODO these two values are taken from a sub-6 GHz cell-free study; check that they are representative of the fronthaul technology assumed here before using them for FR3 numbers.**

a) *Fronthaul rate per split*: What $R_{\text{FH},l}$ contains is set by the functional split of Section IV-E. Let b_{FH} be the number of bits per real sample on the fronthaul, so a complex sample costs $2b_{\text{FH}}$ bits. Under S1 the link carries M streams of frequency-domain samples at the constellation rate $f_{s,\text{I}}$, in both directions,

$$R_{\text{FH},l}^{\text{S1}} = 2b_{\text{FH}} M f_{s,\text{I}}, \quad (86)$$

which becomes $2b_{\text{FH}} M f_{s,\text{II}}$ if the CPU also performs the OFDM modulation and time-domain samples are forwarded. Under S2 the link carries the payload of the served users plus

the precoding coefficients, an $M \times K_l$ complex matrix per subcarrier refreshed once per coherence block of v_{coh} samples,

$$R_{\text{FH},l}^{\text{S2,DL}} = \sum_{k \in \mathcal{D}_l} R_{\text{DL},k} + \frac{f_{s,\text{I}}}{v_{\text{coh}}} 2b_{\text{FH}} M K_l. \quad (87)$$

Under S3 only the payload crosses in the downlink, $R_{\text{FH},l}^{\text{S3,DL}} = \sum_{k \in \mathcal{D}_l} R_{\text{DL},k}$. In the uplink both S2 and S3 forward K_l streams of per-AP partial sums rather than M streams of raw samples,

$$R_{\text{FH},l}^{\text{S2,UL}} = R_{\text{FH},l}^{\text{S3,UL}} = 2b_{\text{FH}} K_l f_{s,\text{I}}, \quad (88)$$

which for S2 costs nothing in rate, by the distributivity argument of Section IV-F. The power-control coefficients are refreshed only once per large-scale fading realisation and are neglected, as in [5]; the precoding coefficients of (87) are not, because they change every coherence block.

The signalling mode deserves separate treatment. Centralized operation requires the CPU to see the raw received pilots in order to estimate the collective channel, so during the uplink signalling phase the link must carry samples even under S2, giving $R_{\text{FH},l}^{\text{sig}} = 2b_{\text{FH}} M f_{s,\text{I}}$ for S1 and S2 alike. Under S3 the pilots are consumed locally and $R_{\text{FH},l}^{\text{sig}} \approx 0$. Centralization therefore imposes a sample-level fronthaul during pilot transmission whatever the data-phase split.

b) *Frame averaging*: The link is bidirectional and stays up across the whole frame, so its static term is charged once per link with a single duty cycle rather than once per direction. Let

$$\theta_l \triangleq \sum_{i \in \{\text{DL}, \text{UL}\}} \tau_i [\tau_{i,\text{sig}} + (1 - \tau_{i,\text{sig}}) \bar{x}_{i,l}] \quad (89)$$

be the fraction of the frame in which the link is in a working mode. With δ_{FH}^μ the reduction factor of an idle transceiver, the frame-averaged fronthaul consumption is

$$\overline{P_{\text{FH},l}} = [\theta_l + (1 - \theta_l) \delta_{\text{FH}}^\mu] P_{\text{FH},0} + \Pi_{\text{FH}} \overline{R_{\text{FH},l}}, \quad (90)$$

with the frame-averaged fronthaul rate obtained by reusing (67) with a zero base level, since an idle link carries no traffic,

$$\overline{R_{\text{FH},l}} = \sum_{i \in \{\text{DL}, \text{UL}\}} \mathcal{A}_{i,l}^{\text{FH}}(R_{\text{FH},l}^i, R_{\text{FH},l}^{\text{sig}}, 0). \quad (91)$$

Optical transceivers wake slowly and hold most of their consumption while idle, so δ_{FH}^μ is close to one in practice and the static fronthaul term is nearly a constant per active link. This is what makes $\mathcal{S}$, rather than the duty cycle, the effective lever on the static part.

Remark 5 (Where the frame structure actually enters). *The map from rate to power in (85) is linear, and $\mathcal{A}_{i,l}^{\text{FH}}$ with a zero base is itself linear in its arguments. For the data-sharing splits S2 and S3 the quantity to insert in (91) is therefore the peak rate during the data mode, and since (28) already delivers $R_{\text{DL},k}$ with the prelog $\bar{x}_i \tau_i (1 - \tau_{i,\text{sig}})$ applied, the downlink data contribution collapses back to $\sum_{k \in \mathcal{D}_l} R_{\text{DL},k}$. Charging the delivered rate directly is thus equivalent to frame-averaging the peak rate, and doing both would count*

the frame structure twice. For the sample-forwarding split S1 the situation is the opposite: $R_{\text{FH},l}^{\text{S1}}$ in (86) is a hardware constant with no dependence on the delivered rate at all, and the frame structure is then the only mechanism that reduces the fronthaul load. The frame structure and the traffic dependence are therefore not two independent knobs; each split makes one of them the active one.

Remark 6 (Fronthaul ceiling on energy efficiency). Under a data-sharing split the same payload is duplicated to every AP of a user's serving cluster, so the network fronthaul load is $\sum_l \sum_{k \in \mathcal{D}_l} R_{\text{DL},k} = \sum_k |\mathcal{L}_k| R_{\text{DL},k}$. Keeping only this term in (64) and using $\sum_k |\mathcal{L}_k| R_{\text{DL},k} \geq (\min_k |\mathcal{L}_k|) \sum_k R_{\text{DL},k}$ bounds the downlink energy efficiency by

$$\text{EE}_{\text{DL}} \leq \frac{1}{\Pi_{\text{FH}} \min_k |\mathcal{L}_k|}, \quad (92)$$

independently of the bandwidth, the transmit power, and the precoder. Full cell-free operation has $|\mathcal{L}_k| = L$, so with $L = 40$ and $\Pi_{\text{FH}} = 0.25 \text{ W per Gbit/s}$ the network cannot exceed 100 Mbit J^{-1} however well everything else is designed. Shrinking the serving clusters raises the ceiling in direct proportion, which is why AP selection is the dominant lever on cell-free energy efficiency and why this term, absent from the co-located model, can decide the comparison on its own.

H. Delivered rate and network energy efficiency

The per-user downlink rate follows from the SINR of (35) inserted into the ergodic rate expression (28),

$$R_{\text{DL},k} = \bar{x}_{\text{DL}} \tau_{\text{DL}} (1 - \tau_{\text{DL},\text{sig}}) \frac{\tilde{B}}{Q} \times \mathbb{E} \left(\sum_{q=1}^Q \log_2 \left(1 + \text{SINR}_k^{\text{dl}}[q] \right) \right), \quad (93)$$

with $R_{\text{DL}} = \sum_{k=1}^K R_{\text{DL},k}$ the delivered sum rate. The per-user decomposition, which the co-located model did not need, is required here because the fronthaul term (87) charges each AP only for the users in its cluster. The network energy efficiency is then

$$\text{EE}_{\text{net}} = \frac{R_{\text{DL}} + R_{\text{UL}}}{P_{\text{net}}} \quad [\text{bit/J}], \quad (94)$$

which reduces to (29) for a single AP without fronthaul. Because a distributed and a co-located deployment cover the same area with different hardware counts, a comparison at equal LM and equal total radiated power is the meaningful one, and the area energy efficiency EE_{net}/A over the served area A is the fairer figure when the deployments differ in extent.

A capacity constraint should accompany (94). If the fronthaul links have a finite capacity C_{FH} , then $R_{\text{FH},l} \leq C_{\text{FH}}$ must hold for every l , and under S1 this is a hard constraint on M and the sampling rate that is independent of the traffic. When it binds, the rate of (93) is not achievable as written and the model returns an optimistic energy efficiency.

I. What the extension does not yet capture

Several effects are left out and are flagged here rather than silently absorbed into the parameters. Fronthaul quantization is modelled only through its rate: the b_{FH} bits per sample of (86) add a distortion term to the effective SINR of (35) that is not present in Section II, so S1 is charged for its fronthaul load but not debited for the accuracy it loses. Channel estimation processing is likewise unpriced: Section II assumes perfect CSI, so neither the $LM \times K$ estimation at the CPU under centralized operation nor the $M \times K$ local estimation under S3 appears in (84) or (77), although both scale with the deployment. The synchronization and reciprocity-calibration overhead is charged as a continuous power P_{sync} but not as frame time, even though over-the-air calibration consumes symbols that would otherwise carry data. Finally, the reduction factors δ_c^μ and δ_c^0 , the supply-and-cooling efficiencies, and the PA parameters are all inherited from a macro base station and are the most likely source of systematic error when they are applied unchanged to a small outdoor AP.

TODO Parameter values still to be sourced before the distributed model can be evaluated. Where a value cannot be pinned down, report results as a sensitivity range over it rather than picking a point value silently.

New parameters, with no counterpart in Section I and therefore no inherited value:

- $\eta_{\text{CPU},\text{sc}}$ and $\eta_{\text{FH},\text{sc}}$ in (64): supply and cooling of the central unit (an inverse data-centre PUE) and of the fronthaul transceivers.
- δ^{ds} , $P_{\text{AP},l}^0$, $P_{\text{FH},l}^0$ in (64): deep-sleep reduction factor and the AP and fronthaul base levels it scales. Only needed once $\mathcal{S} \neq \{1, \dots, L\}$.
- P_{sync} in (76): per-AP distribution of the frequency and phase reference, and reciprocity calibration.
- $P_{\text{CPU},0}$ in (84): always-on consumption of the central unit. The number of FPGAs it runs the centralized precoder on is also still implicit, since (83) names $\bar{M}_{\text{RF}}$ only for the AP side.
- $P_{\text{FH},0}$ and Π_{FH} in (85): currently borrowed from [5]; confirm or replace for the fronthaul technology assumed here.
- b_{FH} in (86): fronthaul quantiser resolution. It sets the S1 fronthaul rate outright and is the largest single lever on it.
- δ_{FH}^μ in (90): idle reduction factor of an optical transceiver, argued to sit near one but not sourced.
- C_{FH} in Section IV-H: fronthaul link capacity, needed before the S1 rates can be claimed feasible.

Inherited values fitted for a macro base station, which should be refitted for AP-class hardware rather than reused:

- ξ , α , $\eta_{\text{PA},\text{max}}$ in (7): small AP amplifiers, not macro-cell ones.
- P_{max} : fix the sizing convention of Remark 2, since $P_{\text{max}} = \rho_{\text{max}}/M$ and a fixed component rating give different scaling laws for the network static floor.

- $\eta_{\text{dig,sc}}$, $\eta_{\text{ana,sc}}$, $\eta_{\text{PA,sc}}$: a passively cooled outdoor AP enclosure is not a macro cabinet, and the cooling share of the overhead largely disappears.
- δ_c^μ and δ_c^0 : micro-sleep and idle reduction factors of AP-class hardware.
- P_s and the 32 chains per FPGA in (83): Remark 4 shows this granularity dominates at small M , so check whether a four-antenna AP really carries a full FPGA at P_s .
- η_{pre} and the other computational efficiencies: the AP-side and CPU-side processors are different device classes, so a single value shared by both nodes is unlikely to hold.
- P_{LO} and $P_{\text{ana,misc}}$: now paid once per AP rather than once per site, so the per-node value matters where the shared one did not.

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